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Editors-in-Chief
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Dear Editors,

We submit our manuscript *TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models* for consideration as a regular paper in the Journal of Machine Learning Research.

Time-series foundation models are now used zero-shot for forecasting, but the interpretability methods developed for language models do not carry over directly. Tokens span different lengths of time in different architectures, real series carry no labels that could validate a probe, and the causal effects of internal features are small enough that uncontrolled measurements mislead. The manuscript presents an open-source system built around these three problems and the results of applying it to seven models.

The main contributions are:

1. **A controlled setting.** A leakage-audited synthetic benchmark with exact generative ground truth, sealed dev and private splits, and a measured token-to-time alignment that puts architectures with tokens of 1 to 96 time steps on a common axis.
2. **Causal concept definitions with matched nulls.** A sparse-autoencoder feature counts as causal only if removing it changes the forecast more than removing the same per-token amount along a random direction on the same series, after reach controls confirm that the intervention lands. A planted-concept forecaster with known answers and decoys holds this test to a fixed false-positive gate (false-positive rate at most 0.003, sensitivity 0.70–1.00).
3. **One-shot confirmation.** Exploratory claims are frozen in a hash-pinned registry and tested once on the sealed split, with multiplicity handled per claim family.
4. **Findings about seven models** (TimesFM, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5, Timer and Time-MoE). Causal features act mainly on forecast level and seasonality. Different architectures select the same inputs for a concept (20 of 20 registered claims confirmed on 971 held-out series, replicated on four private draws), while their causal machinery is organized by model: no concept spans more than three of the seven models. Activation prominence is a poor guide to causal importance (83.5% of prominent features have no causal effect, held out), and depth-resolved reaction profiles replicate for six of nine corruptions.

We believe the paper suits JMLR because its contribution is methodological as much as empirical: it shows that interpretability claims about forecasting models need known answers, matched nulls and held-out confirmation, and it supplies a tool that makes these the default. All code, the benchmark generators, a complete seven-model example run with its held-out confirmation, and a reproduction command for every reported number are public under the MIT license at `https://github.com/`

`DanK10097110/TSFM-Interp`, archived at doi:10.5281/zenodo.23267463 and installable from PyPI.

The manuscript is not under review elsewhere. A preprint is available on arXiv at [arXiv identifier]. We plan to describe the software itself separately in a short paper for the JMLR Machine Learning Open Source Software track; that paper covers the toolkit's design and availability, not the scientific results reported here.

Use of AI tools: Claude (Anthropic) wrote most of the repository's code and tests, ran the experiments and helped draft and edit the manuscript, under the authors' direction. The research questions, ideas and general design choices are the authors'. The authors reviewed the implementation details Claude proposed, checked every reported result and take full responsibility for the content.

Suggested action editors: [names]. Suggested reviewers: [names, affiliations, emails]. We have no conflicts of interest to declare.

Sincerely,

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